CE / CS / SE 3354 · SOFTWARE ENGINEERING · FALL 2026

Project Deliverable P1

Proposal and Requirements — Team Template

Weight 4% · Due Sunday 18 October 2026, 11:59 PM Central · Submit one PDF per team on Canvas

Field	Your entry
Project title	UT Dallas Lost and found system
Team number and section	Team 5 Section 5
Team members (name · net ID)	Michael Makhlouf-mrm230010, Jacob Zepeda-dal637778, Dylan Henson-dal230038, Kellen Widhelm-kcw230001, Jin Jeong-sxj190081, Muneer Bradost-dal176525, Jay Esquivel Savage-jxe240003
Repository URL	https://github.com/ElectronCarrot/SE-Lost_And_Found
Document version	0.1
Date	10/18/2026

0 Revision History and Contribution Record

Both tables below are graded. The contribution record carries the individual 20% of this milestone and must be consistent with your repository history.

Version	Date	Author	Summary of change
0.1	10/9/26		Initial baseline for P1

0.1 Individual contribution record

Team member	What this person produced for P1	Evidence in the repository
Michael Makhlouf	Section 2.1, 2.3, 2.4, 2.5, 3.5, 4.4	
Jacob Zepeda	Section 1.3 (Definitions), presented the project idea to the team	
Dylan Henson	Sections 1.2, 2.2, 3.4, 4.1, 4.3	
Jin Jeong	Section 3.1, 3.2, 3.3	
Kellen Widhelm	Sections 2.3, 4.1	
Muneer Bradost	1.1, 1.2, 1.3, 3.1, 1.4, 4.2	
Jay Esquivel Savage	Completed Section 5 (Open Questions)	

1 Introduction

1.1 Purpose of this document and of the system

This document is a proposal for a UTD Lost and Found website. In this proposal, we will highlight what problems this project will solve, the scope of the project, constraints, assumptions, feasibility, and detailed requirements. This proposal is for the UTD police department which oversees the campus lost and found.

This project is meant to help students and faculty of UTD find their lost items more efficiently. With this website, they will be able to search for their lost item and see if it has been given to the lost and found. This allows them to check without having to physically go to the location.

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
A user interface for UTD students and faculty	Creating student, faculty, and lost and found employee accounts: Every student and faculty member of UTD already has an associated school account.

In scope — we will build this	Out of scope — we will not, and why
A data base for inputting what is found	Not making an app version. To keep it in ab reasonable time frame we will start by making it a website.
An interface for the police department to input what was found	Delivering the items outside of the police station. You will have to go pick them up yourself. We are just making a way to see if we have what you lost and other features.
A way for student and faculty to report and search in the data base, for something missing	We are only making it accessible to UTD students and faculty. We do not want people who have never been to the university to claim items that are not theirs.

Only people attending or working at UTD will have access, so there is no need to create another account when they already have one with the university. Users will likely not want to download an app for such a simple function.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
Match	A link between a lost report and a found item
Lost Report	A student's or faculty's report of an item they are missing
Found Item	An item found by someone on campus which is now being stored at the police station
UTD NetID	Existing university account that is used to login
Seeker	A user that has lost an item and is looking for it

1.4 References

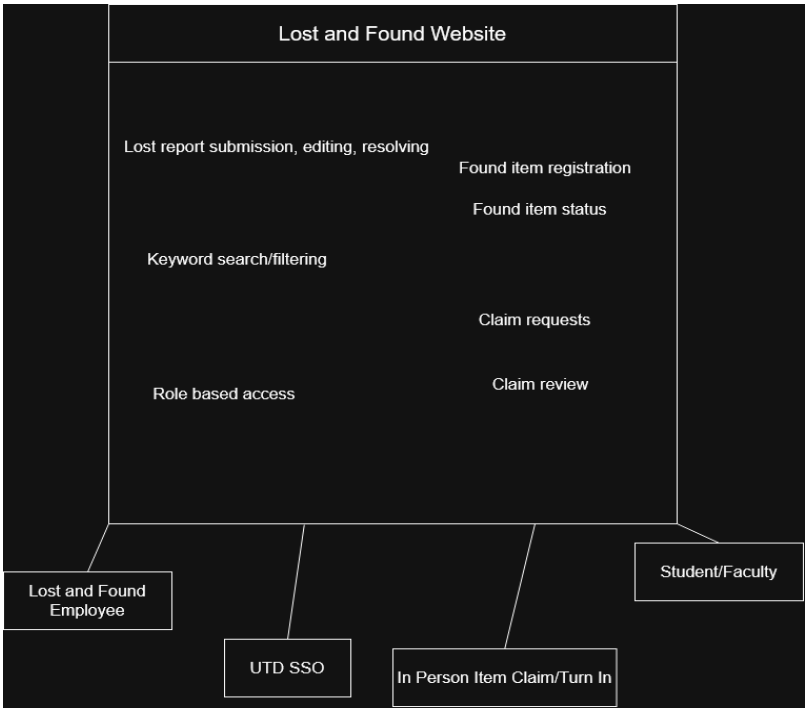
[R1] University of Texas at Dallas. UTD website and university information. https://www.utdallas.edu/

[R2] University of Texas at Dallas. University account and login information (Single Sign-On).

[R3] ISO/IEC/IEEE 29148. Requirements Engineering standard, used as a guide for organizing this document.

2 Overall Description

2.1 Product perspective and context



Everything inside the boundary involves systems we must design for this website, while everything outside the boundary is something that we will rely on. We will be using the UTD single sign on service, and we will be relying on the already existing in person item claim/turn in service. Our website will help supplement the process but not replace it.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
Students/faculty	seeker	Friendly interface	We assumed
Lost and found Employee	user	Able to input items the data base	We assumed
UTD Police budget keeper	Owner/client	Lost operations cost	We assumed

2.3 User characteristics

Student or Faculty: A student or faculty member of UTD that has lost an item. They cannot be assumed to already know how to use the website, so the user interface must be simple to understand and walk them through the process. They could be using a mobile phone or a laptop, or desktop computer.

Lost and Found Employees: An employee working for the UTD police department lost and found. They will use a desktop computer for work. Employees will be expected to use the system daily and to be taught how to use it by another employee.

2.4 Constraints

- We will use the existing campus single sign-on service. All our users should already have an existing UTD account, this also provides additional security which prevents outsiders from using the website.
- Users need an internet connection, and a web browser. To access the website, users must have these.
- Photos stored without using a third party service

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	Students/Faculty and Employees already have a UTD SSO	These users will not be able to access the website
A2	Employees of lost and found already have an in person claim process	Users will not be able to claim their items
A3	Found items are physically held at the UTD police station	Students/Faculty will not be able to claim their items in person

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict
Technical	Feasible	We are making a website and database with simple functionality.
Operational	Feasible	Students on campus want an easy way to track down their lost items.
Schedule	Feasible	A basic version can be finished as long as we prioritize essential functions.
Economic / resource	Feasible	Development costs are low, as we are developing it ourselves. There would be a low cost for database services.
Legal	Feasible	Feasible, as long as the website follows privacy laws and university policies

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-01	The system shall authenticate students, faculty, and Lost and Found employees through UTD Single Sign-On.	MUST	Students/faculty; Lost and Found employees	A tester with a valid UTD account can sign in, while a tester without valid UTD credentials cannot access authenticated functions.
FR-02	The system shall allow a student or faculty member to submit a lost report containing an item category, description, date lost, and last-known location.	MUST	Students/faculty	A tester can submit a report only after completing all four required fields, and the report appears in the user’s report list.
FR-03	The system shall allow a student or faculty member to edit an unresolved lost report that they created.	SHOULD	Students/faculty	The report owner can change a field and see the saved change after reloading the report.

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-04	The system shall allow a student or faculty member to mark their lost report as resolved.	SHOULD	Students/faculty	The report owner can mark a report resolved, after which it no longer appears among active reports.
FR-05	The system shall allow a Lost and Found employee to register a found item with its category, description, date found, storage location, and public photograph.	MUST	Lost and Found employees	A tester using an employee account can enter all required fields and retrieve the saved item record.
FR-06	The system shall allow a Lost and Found employee to update the status of a found item to available, claimed, or disposed.	MUST	Lost and Found employees	Each permitted status can be selected and remains saved after the record is reloaded.
FR-07	The system shall allow authenticated users to search available found items by keywords.	MUST	Students/faculty	A search using a word contained in a test item’s public description returns that item.
FR-08	The system shall allow authenticated users to filter available found items by category, date range, and found location.	MUST	Students/faculty	Applying each filter returns only test records that satisfy every selected filter.
FR-09	The system shall display the public details of a found item without displaying its storage location or claimant information.	MUST	Students/faculty; UTD Police	A student account can view the public description, category, date, general found location, and photograph but cannot view restricted fields.
FR-10	The system shall allow a student or faculty member to submit a claim request for an available found item.	MUST	Students/faculty	A tester can submit one claim request for an available item and receives a visible confirmation.
FR-11	The system shall allow a Lost and Found employee to review pending claim requests for each found item.	MUST	Lost and Found employees	An employee account can open a found-item record and see every pending claim request associated with it.
FR-12	The system shall allow a Lost and Found employee to record the final disposition of a claim request as approved or rejected.	MUST	Lost and Found employees	An employee can select either outcome, and the selected outcome remains saved after reloading.

3.2 Quality requirements

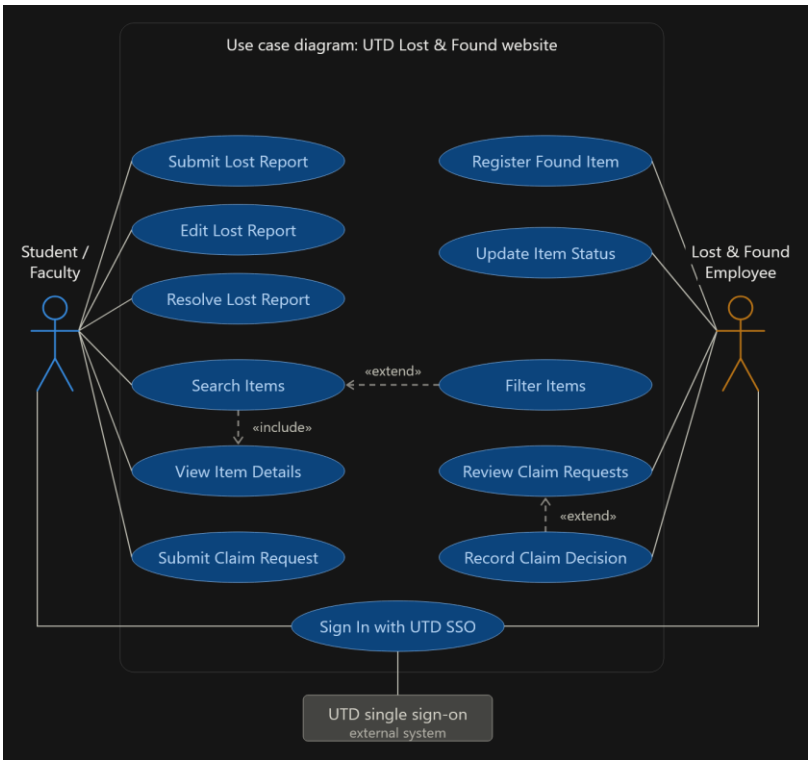
ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	At least 95% of search requests shall display results within 2 seconds under a load of up to 50 concurrent users.	MUST	Students/faculty	Execute 100 timed searches during a 50-user load test and examine the recorded response times.
QR-02	The system shall maintain at least 99% availability during the published service hours, measured over each calendar month and excluding announced maintenance.	SHOULD	UTD Police	Review automated uptime-monitoring records for a one-month test period.
QR-03	A first-time user shall be able to submit a complete lost report within 3 minutes without assistance.	SHOULD	Students/faculty	Conduct a usability test with at least five representative users and time each successful submission.

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-04	Every page used by students, faculty, or employees shall satisfy WCAG 2.2 Level AA criteria evaluated by an automated accessibility scanner and manual keyboard-navigation test.	MUST	Students/faculty	Scan all user-facing pages and manually complete each primary workflow using only a keyboard.
QR-5	The system shall prevent a student or faculty account from accessing another user's non-public report or claim data in 100% of authorization test cases.	MUST	UTD Police	Run positive and negative authorization tests for every protected endpoint and confirm that all unauthorized requests are denied.
QR-6	The system shall preserve 100% of successfully submitted lost reports, found-item records, and claim decisions after an application restart during persistence testing.	MUST	UTD Police	Create a predefined test dataset, restart the application, and compare every stored record and field with the original dataset.

3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The system shall use the university-approved UTD Single Sign-On service for user authentication.	Section 2.4; UTD Police	The team does not control university identities, and separate account creation is outside the agreed scope.
CON-02	The system shall operate through a standards-compliant web browser without requiring installation of a native mobile or desktop application.	Section 1.2; Section 2.4	Native application development is explicitly outside the P1 project scope.
CON-03	The system shall permit access to authenticated functions only while the user has an active internet connection.	Section 2.4	The application depends on the remote authentication service and central database.

3.4 Initial use cases



Field	Content
Identifier and name	UC-01 **submitting** claim to an item
Primary actor	Student/Faculty
Supporting actors	Lost and found Administrator
Preconditions — what must be true before it starts	An item must exist in the database and can be searched for, student/faculty is signed in
Trigger — what starts it	When the Student/Faculty finds an item in the database and summits a claim that it's theirs
Main success scenario — numbered steps	1-search database 2- submits claim 3-claim verified 4-listings altered 5-item picked up
Alternative and exception flows	1-student submits claim 2- claim denied 3-listing stay same
Postconditions — what is true when it succeeds	The lost and found admin reviews a claim
Requirements realised	FR- a way to review claims

Field	Content
Identifier and name	UC-02 Reporting a lost item
Primary actor	Student/Faculty
Supporting actors	Lost and Found Administrator

Field	Content
Preconditions — what must be true before it starts	There must be a lost item
Trigger — what starts it	A student/faculty submits a lost item claim
Main success scenario — numbered steps	1-report a missing item 2-search the database and find do not find the item 3-report is seen by system 4- admin put into data base the missing item 5- notification sent to user
Alternative and exception flows	1-report a missing item 2-search the database and find item 3-claim lost item
Postconditions — what is true when it succeeds	The user is notified that the item has been put in the lost and found
Requirements realised	FR- a way so check if there is a lost report submitted for the item that is being entered to the data base

3.5 Requirements to use case traceability

Requirement ID	Use case(s) that realise it	If blank — why, and what you will do
FR-01	Sign in with UTD SSO	
FR-02	Submit lost report	
FR-03	Edit lost report	
FR-04	Resolve lost report	
FR-05	Register found item	
FR-06	Update item status	
FR-07	Search items	
FR-08	Filter items(extends Search items)	
FR-09	View item details(included by Search items)	
FR-10	Submit claim request	
FR-11	Review claim requests	
FR-12	Record claim decision(extends Review claim requests)	

4 Project Plan

4.1 Team roles and working agreement

Team member	Role for P1 to P2	How the team makes a decision when it disagrees
Dylan Henson	P1- working to get communication started and 3.4 done	Communication to find the way we want to answer the problems.
Michael Makhlouf	System design/diagrams	Votes for major decisions
Jacob Zepeda	Requirements and terminology, keeping 1.3 consistent with Section 3	Discuss as a team, then majority vote
Muneer Bradost	Mainly working on section 1.2, 1.4 and 4.2 QA report prior to submission.	Communicate with group and get to greed version by majority of votes

Team member	Role for P1 to P2	How the team makes a decision when it disagrees
Kellen Widhelm	Work on designing the user experience for submitting lost item report	Talk as a group to come to a conclusion
Jin Jeong	Requirements analysis and documentation, including functional, quality, and constraint requirements; maintain requirements traceability for P2.	Discuss the issue as a team and use a majority vote if a consensus cannot be reached.
Jay Esquivel Savage	Worked on Section 5 open questions, identifying project risks and assumptions, and helping resolve unanswered questions for P2	Discuss the issue as a group and use a majority vote if needed

4.2 Plan to P2

Date	What will be finished	Owner
October 23, 2026	Review the P1 requirements and make changes if needed.	Muneer Bradost
October 26, 2026	Check the functionality and quality requirements to make sure they are clear and testable.	Dylan Henson
October 29, 2026	Review the use cases and make sure they match the requirements.	Jacob Zepeda
November 1, 2026	Work on the system design and diagrams needed for P2.	Michael Makhlouf
November 4, 2026	Check the requirements and diagrams to make sure everything matches.	Jin Jeong
November 7, 2026	Put the final P2 document together and review it as a team before submission.	Kellen Widhelm

4.3 Risks

#	Risk	Likelihood	Impact	Mitigation — action and owner
R1	Slowing of the data base	low	high	Make the date a item can be in the data base lower so getting more out
R2	Data base failure	low	high	Making backups
R3	Duplicate entries	medium	medium	Implementing a way so that when you input a item you make sure that its not a dup
R4	Problems integrating SSO	high	high	Getting help from others that have made similar websites for UTD
R5	Project changing later on	low	high	Make sure we think what could change later on

4.4 Repository and tooling

Repository: https://github.com/ElectronCarrot/SE-Lost_And_Found

Branching: Each task gets its own branch created from the latest main, branches should have meaningful names like “feature/keyword-search”, changes to main must be approved by at least 2 other teammates, branches should be deleted after successfully merging.

Commits: Commits should have meaningful names like “fix: block duplicates”. Each commit should be focused on one key change/fix/feature and not be a combination of multiple, commit often and always test before committing to main.

Issues: Issues should be used to plan ahead, where possible give the issue an owner so it's clear who needs to work on what, where possible issues should be tied to their related requirement/use case, issues should have meaningful names.

Repository owner: Michael Makhlouf

Access: Everyone in the team will have equal write access, Instructor/TA will have read access, whether by making the repository public or by adding them manually.

4.5 Declaration of AI tool use

Tool	What it was used for	Who used it	How the output was checked
OpenAI Codex	Assisted with drafting and revising Section 3 requirements.	Jin Jeong	The team reviewed each suggestion against the project scope and rubric and edited the final wording together.

5 Open Questions

#	The question	Who can answer it	Needed by	What you assumed in the meantime
Q1	Will UTD allow our website to use the existing university Single Sign-On system?	UTD IT Department	10/23	We assume students, faculty, and employees can use their existing UTD accounts to log in.
Q2	How will the police department verify that someone actually owns a lost item before giving it back?	UTD Police Department	10/23	Employees will verify ownership in person before returning an item.
Q3	Who will be responsible for updating the database when items are found, claimed, or disposed of?	UTD Lost and Found Employees	10/26	Lost and found employees will be responsible for keeping item information updated.

6 Baseline Declaration

We confirm that the requirements in section 3 represent our agreed baseline as of the submission date, and that every team member ,who has signed, read this document in full.

Team member	Signature or typed name	Date
Dylan Henson	DH	10/18/2026
Michael Makhlouf	Michael Makhlouf	10/18/2026
Jacob Zepeda	JZ	10/18/2026
Muneer Bradost	Muneer Bradost	10/18/2026
Kellen Widhelm	Kellen Widhelm	10/18/2026
Jin Jeong	Seongjin Jeong	10/18/2026
Jay Esquivel Savage	JES	10/18/2026

How P1 Is Marked

P1 is worth 4% of the course grade. Following the syllabus, 80 points are for the shared team artifact and 20 for individual contribution evidence.

Criterion	Points	What full marks look like
Problem, scope and context (1.1, 1.2, 2.1)	10	The problem is a problem somebody actually has. The out-of-scope column is filled and the exclusions are justified. The context diagram's boundary matches what sections 2 and 3 claim.
Stakeholders and elicitation (2.2, 2.3)	10	Stakeholders include at least one who will never use the system. Every need has a stated basis, and assumptions are labelled as assumptions rather than disguised as findings.
Requirements quality (3.1, 3.2, 3.3)	25	Every requirement is one testable “shall” sentence with an identifier, a priority, a source and an acceptance criterion. Quality requirements carry a threshold, a condition and a method. Constraints are distinguished from quality requirements.
Use cases and traceability (3.4, 3.5)	20	Use cases are at actor-goal level, not steps. Two are written out in full, including alternative flows. The matrix is complete and the team says what the gaps mean and what they did about them.
Feasibility and plan (2.6, 4)	10	Each feasibility verdict is backed by evidence a reader can check. The plan has dates and owners. Risks are future events with actionable mitigations.
Open questions and baseline (5, 6)	5	Real open questions, with an owner and a date, and an honest statement of what was assumed in the meantime.
Document quality	—	Guidance boxes removed, sections in order, within the length budget, readable. Marks are deducted here, not awarded.
Individual contribution evidence (0.1)	20	Every member's contribution is distinguishable and traceable to something in the repository a reader can open.